Biology

Topic: Biological Classification

Introduction: Kingdoms of Life

1. Consider the following statements with respect to the characteristic features of the five-kingdom system.
(a) In Animalia the mode of nutrition is autotrophic.
(b) In Monera the nuclear membrane is present.
(c) In Protista the cell type is prokaryotic.
(d) In Plantae the cell wall is present.
Of the above statements:
(1) Of the above statements:
(2) (b) and (d) are correct
(3) (c) and (d) are correct
(4) (d) alone is correct

2. Two kingdom classification of Linnaeus cannot distinguish between
(1) Unicellular and multicellular organisms
(2) Eukaryotes and prokaryotes
(3) Photosynthetic and non-photosynthetic organisms
(4) All of these

3. Classification suggested by R. H. Whittaker is not based on
(1) Cell structure
(2) Reserve food material
(3) Reproduction
(4) Body Organisation

4. Read the following assertion and reason statements and select the correct option:
Assertion: Two kingdom classification system used for a long time was found inadequate.
Reason: A large number of organisms did not fall into either category (i.e., plants and animals).
(1) Both assertion and reason are true, and reason is the correct explanation of the assertion
(2) Both assertion and reason are true, but reason is not the correct explanation of the assertion
(3) Assertion is true, but reason is false
(4) Both assertion and reason are false

5. The number of kingdoms in five kingdom classification with only chemoautotrophic mode of nutrition is/are
(1) One
(2) Three
(3) Two
(4) Four

6. In five kingdom classification, cellular level of body organisation and lack of nuclear membrane are the characteristic of
(1) Kingdom Monera
(2) Kingdom Animalia
(3) Kingdom Protista
(4) Kingdom Fungi

7. According to Whittaker's five kingdom classification, Chlorella and Chlamydomonas are kept along with Amoeba and Paramoecium under
(1) Monera
(2) Protista
(3) Plantae
(4) Algae

8. Over the time, an attempt has been made to evolve a classification system which reflects
(1) Not only the morphological, physiological and reproductive similarities, but also phylogenetic relationships.
(2) Only the morphological and reproductive similarities.
(3) Only the morphological and physiological similarities.
(4) Only morphological similarities but not physiological and reproductive similarities.

9. Read the following statements and select the correct option:
Statement I: In earlier classification, prokaryotic bacteria and the blue green algae (cyanobacteria) were placed together with other groups which were eukaryotic.
Statement II: The two-kingdom classification did not differentiate between the heterotrophic group - fungi, and the autotrophic green plants.
(1) Statement I is true, but statement II is false.
(2) Both statement I and statement II are false.
(3) Both statement I and statement II are true.
(4) Statement I is false, but statement II is true.

10. Which one of the following is incorrect?
(1) R.H. Whittaker proposed a five-kingdom classification and defined by him were named Monera, Protista, Fungi, Plantae and Animalia.
(2) In Linnaeus time, two kingdom system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively.
(3) The two-kingdom classification has put together organisms which, in earlier classifications, were placed in different kingdoms.
(4) The main criteria for five kingdom classification includes cell- structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.

Kingdom Monera

11. Fill in the blanks and choose the correct option:
(I) ___ (A) _____ are the sole members of kingdom Monera.
(II) Majority of bacteria show ___ (B) ____ mode of nutrition.

	A	B
(1)	Amoeba	Saprobic
(2)	Bacteria	Heterotrophic
(3)	Bacteria	Chemoautotrophic

(4)	Protozoans	Holozoic

12. Read the following statements and choose the correct option.
Statement I: Bacteria are simple in structure and complex in behaviour.
Statement II: Bacteria shows the most extensive metabolic diversity.
(1) Only statement (1) is correct
(2) Only statement (II) is correct
(3) Both statements (1) and (II) are correct
(4) Both statements (1) and (II) are incorrect

13. Bacteria reproduce mainly by
(1) Transformation
(2) Fission
(3) Fragmentation
(4) Budding

14. The Archaebacteria which occur in gut of ruminant animals
(1) Are responsible for production of biogas
(2) Are also present in marshy habitat
(3) Are called methanogens
(4) All of these

15. Archaebacteria differ from eubacteria in
(1) Mode of reproduction
(2) Cell membrane structure
(3) Mode of nutrition
(4) Cell shape

16. Consider the following statements and state true (T) and false (F):
(A) Bacteria cannot live in snow and deep oceans.
(B) Halophiles are found in marshy habitat and are nutritionally autotrophs.
(C) Under unfavourable conditions bacteria produce several types of spores.
(D) Bacteria show autotrophic and heterotopic nutrition.

	A	B	C	D
(1)	T	F	F	T
(2)	F	F	F	T
(3)	F	F	T	T
(4)	T	T	F	F

17. Read the following assertion and reason statements and select the correct option:
Assertion (A): Cyanobacteria are photosynthetic blue - green algae with all the prokaryotic structures.
Reason (R): They are green due to the presence of chloroplasts.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, and R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

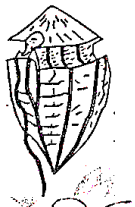
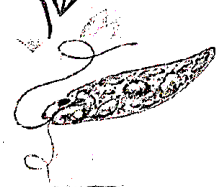
Kingdom Protista

18. Choose the odd one out w.r.t. features of Protists:
(1) Boundaries of this kingdom is not well-defined.
(2) Cell structure is eukaryotic.

(3) Show locomotion by cilia or flagella.
(4) Largely terrestrial in habitat

19. Read the following statements about ciliated protozoans and select the correct option:
Statement I: They have a cavity (gullet) that opens to the outside of the cell surface.
Statement II: The coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet.
(1) Statement I is true, but statement II is false.
(2) Both statement I and statement II are false.
(3) Both statement I and statement II are true.
(4) Statement I is false, but statement II is true.

20. Which of the following is incorrect about chrysophytes?
A. Found in fresh water as well as in marine environments.
B. Have indestructible cell wall embedded with silica.
C. Can float passively on the surface of water.
D. Absence of asexual reproduction.
(1) A, B, C and D
(2) D only
(3) B and C
(4) C and D

21. Read the following statements:
(a) Mostly marine and photosynthetic.
(b) Appear yellow, green, brown, blue or red depending on the main pigments present in their cells
(c) Has two flagella lying perpendicular to each other.
The above statements are assigned to
(1) 
(2) 
(3)
(4)

22. Identify the organisms on the basis of given information and select the correct option:
(I) The cell wall has stiff cellulose plate on the other surface.
(II) Undergo rapid multiplication and make the sea appear red.
(1) Euglenoids
(2) Gonyaulax
(3) Diatoms

(4) Trichodesmium

23. Ciliated protozoan is

(1) Paramoecium

(2) Plasmodium

(3) Trypanosoma

(4) Entamoeba

24. Match the columns and choose the correct option.

Column I		Column II	
A	Euglenoids	(i)	Red tide
B	Gonyaulax	(ii)	Saprophytes
C	Chrysophytes	(iii)	Stagnant freshwater
D	Slime moulds	(iv)	Golden algae

(1) (A) - (ii), B - (i), C - (iv), (D) - (iii)

(2) (A) - (ii), B - (iv), C - (i), (D) - (ii)

(3) (A) - (iii), B - (i), C - (iv), (D) - (ii)

(4) (A)- (ii), B - (iii), C - (i), (D) - (iv)

25. Spores possess true walls and are extremely resistant and survive for many years even under adverse condition in

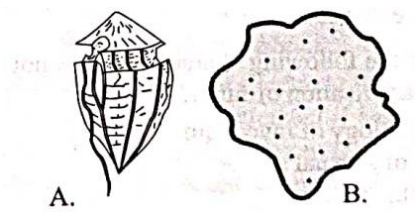
(1) Sporozoans

(2) Euglena

(3) Sarcodines

(4) Slime moulds

26. Identify the organisms shown in the figure and select the similarity in features between them:



(1) Nature of cell wall

(2) Body organisation and cell type

(3) Mixotrophic mode of nutrition

(4) Habitat and sexual reproduction

27. Red tide forming members

(1) Have two similar-sized flagella in the apical part.

(2) Have stiff cellulosic plates on the inner surface of cell wall.

(3) Release toxins which can kill other marine animals.

(4) More than one option is correct

Kingdom Fungi

28. All are the characteristic features of fungi, except

(1) Chitinous cell wall

(2) Cosmopolitan occurrence

(3) Filamentous thalloid body

(4) Autotrophic organisms

29. The cell wall of fungi consists of
(1) Chitin
(2) Cellulose
(3) Hemicellulose
(4) Lignin

30. Heterotrophic filamentous organisms belongs to
(1) Kingdom Monera
(2) Kingdom Protista
(3) Kingdom Fungi
(4) Kingdom Animalia

31. Read the following assertion and reason statements and select the correct option:
Assertion (A): All fungi are filamentous.
Reason (R): Yeasts are multicellular.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, and R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R is false

32. Read the following statements w.r.t. fungi:
A. The fungi constitute a unique kingdom heterotrophic organisms.
B. They prefer to grow in dark, warm and humid places.
C. They show a great diversity of morphology and habitat.
(1) Only (A) is correct.
(2) Only (A) and (B) are correct.
(3) Only (A) and (C) are correct.
(4) All are correct.

33. Which of the following characteristics is not a criterion for the classification of fungi?
(1) Morphology of mycelium
(2) Type of asexual spores
(3) Fruiting bodies
(4) Reserve food materials

34. Which of the following is a unicellular fungus?
(1) Saccharomyces
(2) Agarics
(3) Penicillium
(4) Mucor

35. Select a correct statement w.r.1. dikaryotic phase in fungi:
(1) It comes after meiosis in asexual reproduction.
(2) It is present in vegetative phase of sac fungi.
(3) It occurs in the life cycle of members of class phycomycetes only.
(4) It occurs when karyogamy is delayed.

36. White spots seen on mustard leaves are due to a fungal disease caused by
(1) Puccinia
(2) Albugo
(3) Synchytrium
(4) Rhizopus

37. Rhizopus, the common bread mould, shows all of the given features, except

(1) Coenocytic mycelium
(2) Asexually reproduced by sporangiospores
(3) Oosphere is formed by fusion of gametes
(4) Possess chitinous cell wall

38. Read the following statements:

i. Has branched septate mycelium.
ii. Produce conidia for asexual reproduction.
iii. Produce non-motile endogenously produced sexual spores.

Identify the class of fungi on the basis of above features:

(1) Ascomycetes
(2) Basidiomycetes
(3) Deuteromycetes
(4) Phycomycetes

39. Members of class phycomycetes are characterized by all of the following features, except

(1) Mycelium is aseptate and coenocytic.
(2) Members could be found in aquatic habitat.
(3) Presence of prolonged dikaryotic phage.
(4) Reproduce asexually by aplanospores or motile zoospores.

40. Match the following column and choose the correct option.

Column I		Column II	
1	Fungi that depend on living plants and animals	a	Karyogamy
2	Fungi in association with algae	b	Lichens
3	Fungi in association with root of higher plants	c	Parasites
4	Fusion of two nuclei	d	Mycorrhiza

(1) 1-c, 2-d, 3-b, 4-a
(2) 1-d, 2- b, 3- c, 4- a
(3) 1-c, 2-b, 3-d, 4-a
(4) 1-a, 2-b, 3-c, 4-d

41. Where does meiosis occur in mushrooms?

(1) Ascus
(2) Basidium
(3) Ascus mother cell
(4) Zygospore

Kingdom Plantae and Animalia

42. Cuscuta is

(1) An insectivorous plant
(2) A total stem parasite
(3) A partial stem parasite
(4) A partial heterotrophic plant

43. The eukaryotic heterotrophic, multicellular organisms which lack cell wall

(1) Are included in kingdom Animalia
(2) Have holozoic mode of nutrition
(3) Digest their food in internal cavity

(4) All ofthese

44. Members of kingdom Animalia differ from that of Plantae in

(1) Being eukaryotic
(2) Being multicellular
(3) Absence of cell wall
(4) All of these

45. Glycogen is a reserve food in all, except

(1) Animals
(2) Bacteria
(3) Plants
(4) Fungi

46. Read the following statements:

Statement A: Few members of kingdom Plantae are partial heterotrophs and parasites.

Statement B: Animals follow a definite growth pattern and grow into adult that have a definite shape and size.

(1) Only (A) is correct
(2) Only (B) is correct
(3) Both (A) and (B) are correct
(4) Both (A) and (B) are incorrect

47. In kingdom Plantae, alternation of generation is not associated with

(1) Sporophytic and gametophytic phases
(2) Length of haploid and diploid phases
(3) Number of haploid and diploid phases
(4) Free living or dependent nature of haploid and diploid phiase

48. Given below are features of members of kingdom Animalia:

i. Most forms are capable of locomotion.
ii. They digest their food in an internal cavity.
iii. Show sexual reproduction by copulation of male and female followed by embryological development.
iv. Reserve food material is similar to plants.
v. Higher and lower forms show elaborate sensory and neuromotor mechanism.

How many statements given above are correct?

(1) Three
(2) Four
(3) Two
(4) Five

49. Bladderwort is a

(1) Parasitic plant
(2) Partial heterotroph
(3) Insectivorous plant
(4) All expect (1)

50. Life cycle of plants has two distinct phases

(1) The haploid sporophytic and the diploid gametophytic
(2) The diploid sporophytic and the haploid gametophytic
(3) Both sporophyte and gametophyte are diploid
(4) Both sporophyte and gametophyte are haploid

51. Read the following statements and select the correct option:
Statement I: Animals are directly or indirectly dependent on plants for food.
Statement II: Plants are photosynthetic, but few members are partially heterotrophic.
(1) Statement I is true, but statement II is false.
(2) Both statement I and statement II are false
(3) Both statement I and statement II are true.
(4) Statement I is false, but statement II is true.

Viruses, Viroids, Prions, Lichens

52. Virus is a/an
(1) Obligate parasite
(2) Facultative parasite
(3) Obligate saprophyte
(4) Facultative saprophyte

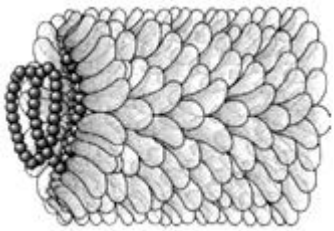
53. How many of the following statements are correct w.r.t. virus?
a. They are obligate intracellular parasites.
b. They are inert outside their specific host.
c. Virus can be crystallized.
d. No virus contains either DNA or RNA.
e. The protein coat called capsid is made up of small subunits.
(1) Five
(2) Four
(3) Three
(4) Two

54. Which of the following symptoms is not related with viral infection in plants?
(1) Mosaic formation
(2) Wilting and tissue death
(3) Leaf rolling and leaf curling
(4) Yellowing and vein clearing

55. Viruses do not have any position in Whittaker's system because of
(1) Infectious nature
(2) Non-cellular structure
(3) Obligate intracellular parasites
(4) Their ability to take over cellular machinery of host for multiplication

56. Read the following assertion and reason statements and select the correct option:
Assertion (A): Viruses are obligate parasites.
Reason (R): No virus contains both RNA and DNA.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true, and R is not the correct explanation of A
(3) A is true but R is false
(4) Both A and R are false

57. Identify the figure and select the incorrect statement



(1) Capsomere are arranged in helical form.
(2) Passes non-infectious RNA.
(3) Crystals consist largely of proteins.
(4) It is the first virus to be discovered.

58. Lichens are symbiotic association between two components which are, respectively
(1) Fungi and roots of higher plants
(2) Cellular and acellular
(3) Living and non-living
(4) Algae and fungi

59. Lichens are dual organisms that
A. Are perennial
B. Grow rapidly
C. Act as pioneer communities in biological succession in soil and water.
(1) Only (A) and (B) are correct
(2) Only (A) and (C) are correct
(3) (B) and (C) are incorrect
(4) All (A), (B) and (C) are correct

60. Select the incorrect statement about viroids:
(1) They were discovered by T. O. Diener.
(2) They are infectious RNA molecule.
(3) They are known to cause potato spindle tuber disease in plants.
(4) They are larger than virus.

61. Lichens are considered as indicators of SO_2 pollution because
(1) They love to grow in SO_2 polluted areas.
(2) They do not grow in SO_2 polluted areas.
(3) They can tolerate air pollution.
(4) They oxidize sulphur to sulphuric acid under aerobic conditions.

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	4	2	1	1	1	2	1	3	3	2	3	2	4	2
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
2	3	4	3	2	1	2	1	3	4	2	3	4	1	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
4	4	4	1	4	2	3	1	3	3	2	2	4	3	3
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
3	3	1	4	2	3	1	2	2	2	2	2	4	3	4
61														
2														